🌻 The factors table

📅 22 Sep 2025

Summary

This extension builds a **factors table** from the (already filtered) **links table**.

- **Input**: Links table (one row per coded causal claim), already filtered by whatever link-level filters are active (date range, source subset, link-type, polarity, etc.).
- **Output**: Factors table (one row per factor label) with optional **in/out splits**, optional **group breakdown columns**, optional **normalised views**, and optional **significance tests**.

The key idea is: **everything starts from links**, and the factors table is just an aggregation of link-derived "factor mentions".

What exactly is a "factor mention"?

Each link row typically contains at least:

- a **cause label** and an **effect label**
- a `source_id` (and source metadata available for grouping)

From each link row we derive **two factor-mention records**:

- one mention for the cause label (direction = `out` / "as cause")
- one mention for the effect label (direction = `in` / "as effect")

These mention records are the atomic units that the factors table aggregates.

How it works (links → factor mentions → factors table)

Step 0. Start from the already-filtered links table

All upstream filters apply first. This filter does **not** try to re-implement link filtering; it only consumes the current link set.

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Step 1. Expand each link row into factor-mention records

For each link row r:

- emit mention $m_{out} = (factor = r.cause, direction = out, source_id = r.source_id, link_id = r.link_id)$
- emit mention $m_{in} = (factor = r.effect, direction = in, source_id = r.source_id, link_id = r.link_id)$

Notes:

- This is why "citation totals across factors" are not "number of links": each link yields (at least) **two mentions**.
- Self-loops can either yield 2 mentions with the same label, or be deduplicated; whichever rule is chosen must be stated consistently in the UI.

Step 2. Apply factor label transforms (rewrite layer)

Before aggregating, apply the current label-rewrite transforms to `m.factor`, such as:

- **Collapse factor labels** (bucket near-synonyms by search term)
- **Exclude bracket text** (e.g. `Education (primary)` → `Education`)

These are **temporary rewrites** for analysis/presentation; they do not change the underlying coding.

Step 3. Aggregate into the base factors table

Group mention records by factor label f. Compute one or more base columns (examples):

- **Citation Count (total)**: number of mention records for f
- **Citation Count (out / in)**: split by `direction`
- **Source Count (total)**: number of distinct sources that mention f at least once
- **Source Count (out / in)**: distinct-source counts split by `direction`

Derived role measures (examples):

- **Outcomeness**: a summary of "effect-like vs cause-like" usage (e.g. based on in/out counts)

The factors table is therefore an **interpretation layer**: it encodes choices about (a) rewrite rules, (b) evidence unit (citations vs sources), and (c) whether direction matters.

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Groups: breakdown columns

Let G be a group variable defined on sources (e.g. `district`, `gender`, `section`). This filter can add **group breakdown columns** by aggregating mention records jointly by $(factor, group)$.

Step 4. Join source metadata to mentions (for grouping)

Join each mention record with its source attributes so each mention has $G(m)$.

Step 5. Add the group breakdown columns

For each factor f and each group level g, compute cells such as:

- **Citations mode**: You can't use 'macro parameter character #' in math mode
- **Sources mode**: You can't use 'macro parameter character #' in math mode

Optional: also split each cell by direction (`in` / `out`) if the UI supports it.

What group columns are for

They let you ask:

- "Which factors are disproportionately mentioned by group A vs B?"
- "Which outcomes differ by district / section?"

Totals and normalisations (what totals actually mean)

Because the factors table is built from factor mentions:

- **Citation totals across factors** are totals of **mentions**, not totals of **links**.
- **Source totals across factors** are totals of **source–factor incidences**, not "number of sources".

Normalisation is a choice of baseline, not a cosmetic option:

- **Within-group totals** (sum over factors within a group column) measure that group's overall mention volume under current filters.
- **Percent-of-baseline view** (within each group) shows *relative prominence*:

The little equation below is just intuition (you can ignore it if you want).

\[\text{share}(f,g) = \frac{\text{cell}(f,g)}{\sum_{f'} \text{cell}(f',g)} \]

This is useful when groups differ in overall verbosity or number of sources.

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Optional inference: significance testing per factor (single grouping variable)

If exactly one group variable G is selected, this filter can compute a per-factor test asking whether mentions for factor f are distributed across group levels differently than expected given group baselines.

Intuition (chi-squared style):

> Even if group A has more mentions overall than group B, is factor f still *over-represented* in one group relative to that baseline?

Examples from the app

Factors table: group differences + tests

Bookmark #535 is a worked example of the factors table with group breakdown and statistical "difference" highlighting.

Showing 1-10 of 170 rows **Page Size** 10 First Prev 1 2 3 4 5 Next Last

Factor	Significant (All)	custom_# Age of the main respondent - 20-45	custom_# Age of the main respondent - 46+	# Citations	# Sources	In-Degree	Out-Degree	Outcomeness	Avg Incoming Sentiment
Increased knowledge	No	158	72	230	18	22	208	0.096	0.000
Farm production	Yes	89	68	157	16	76	81	0.484	0.000
Health behaviour	No	93	58	151	17	26	125	0.172	0.000
Improved health	No	89	54	143	17	136	7	0.951	0.000
Diet improved	Yes	78	23	101	18	81	20	0.802	0.000
Income	Yes	44	44	88	15	47	41	0.534	0.000
Planted new crop/vegetabl	No	42	20	62	13	28	34	0.452	0.000
Improved/new farming tec	Yes	20	24	44	14	21	23	0.477	0.000
Ability to buy food	Yes	14	23	37	12	17	20	0.459	0.000
Food consumption quantit	No	22	15	37	17	31	6	0.838	0.000

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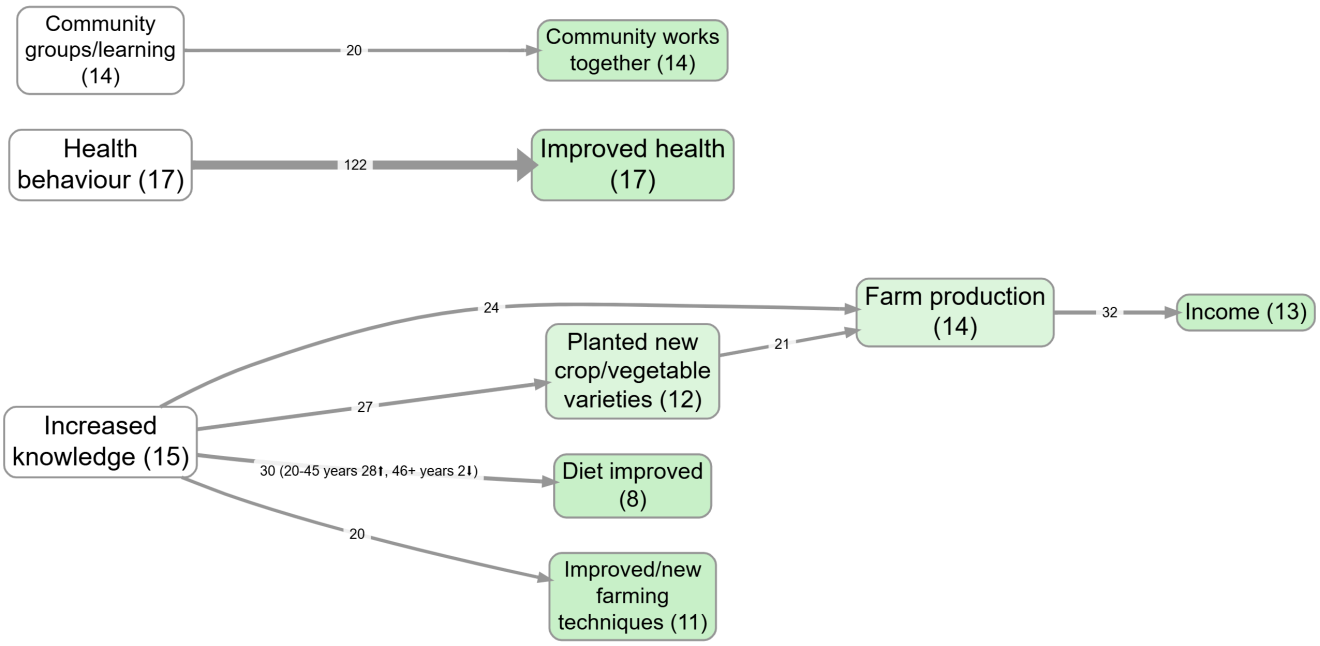
Map: surfacing group differences on links

Group comparisons can also be pushed back onto the map by labelling/colouring link bundles with group information.

Bookmark #980 is an example where link labels summarise group differences.



For ordered groupings (e.g. age bands), an ordinal/trend framing can be more appropriate than treating levels as unordered categories.

Why this is useful

This helps you move from “what is mentioned most?” to “what differs by context/group?”.

- **Everything** (factor counts, roles, group comparisons, tests) is derived from the same link-derived mention records.
- Every result depends on the current link filters and label rewrite transforms.

Examples (from the app)

Factors table: group differences + significance test

Bookmark #535

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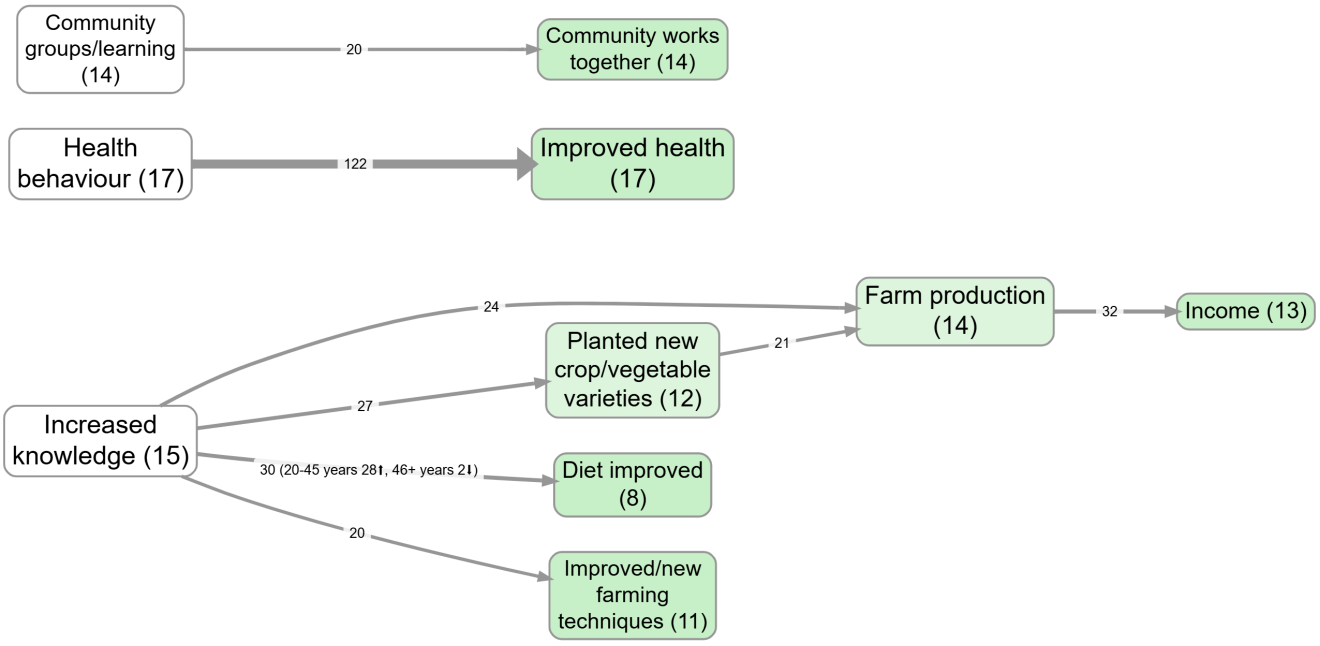
Bringing group differences onto the map (as link labels)

Bookmark #980



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